

Aman Murari Singh

AI Researcher & Machine Learning Engineer

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EDUCATION

B.Tech. in Computer Science and Engineering, Siksha 'O' Anushandhan University 2024 – 2028, CGPA: 8.71

EXPERIENCE

• AI Engineer | September 2025 – June 2026

Doubt Buddy

- Engineered Datagentz, an AI-powered Microsoft Excel add-in that transforms raw spreadsheets into actionable insights through natural language queries, automated pattern detection, and smart visualizations.
- Built AI workflows for spreadsheet understanding, PDF-to-data extraction, data transformation, conversational analytics, and smart visualizations inside Excel.
- Integrated AI capabilities into Excel's native environment, enabling non-technical users to perform advanced data analysis without writing complex formulas.
- Worked on end-to-end AI product development, including feature design, backend integration, data workflow optimization, and Microsoft Marketplace publication.
- Product Link: <https://marketplace.microsoft.com/en-us/product/WA200009638>

• AI Research Intern | May 2025 - October 2025

Vizuara Technologies Private Limited

- Worked on nano-GPT-OSS, an open-source research project focused on efficient Transformer architecture design and small-scale language model training.
- Researched and implemented hybrid attention mechanisms combining sliding-window attention and sink tokens to improve model efficiency and context handling.
- Integrated MoE-enhanced MLP blocks to improve model capacity while maintaining memory and storage efficiency.
- Achieved 12-30% lower validation loss compared to GPT-2 across 6-12 layer model configurations.
- Reduced RAM usage by 11% and disk footprint by 11%, while maintaining stronger model performance despite a slightly larger parameter count.
- Demonstrated improved scaling behavior with a 30.5% validation loss improvement at 12 layers, showing better generalization at larger depths.
- Implemented distributed training using PyTorch Distributed Data Parallel, improving scalability for multi-GPU language model pretraining.
- Open-sourced code, training logs, and evaluation metrics, helping the repository reach 100+ GitHub stars and 30+ forks.
- GitHub: <https://github.com/VizuaraAI/nano-gpt-oss>

PROJECTS

– Datagentz: AI-Powered Excel Add-in

AI assistant for MS Excel spreadsheet analysis, PDF-to-data extraction, smart visualization, and natural language data workflows.

- * Developed an AI-powered Excel add-in that allows users to ask questions about spreadsheet data, extract insights, transform data, and generate visualizations using natural language.
- * Built features for conversational data analysis, automated pattern detection, PDF-to-data extraction, and spreadsheet-based AI workflows.
- * Focused on making advanced analytics accessible for non-technical users by reducing dependency on manual formulas and complex Excel operations.

– nano-GPT-OSS: Efficient Transformer Research

Open-source research project focused on efficient Transformer variants for language model training.

- * Implemented hybrid attention using sliding-window attention and sink tokens to improve efficiency in small-scale language models.
- * Added MoE-enhanced MLP blocks to improve model capacity and parameter efficiency.
- * Benchmarked models against GPT-2 baselines using validation loss, RAM usage, disk footprint, and scaling behavior.
- * Achieved 12-30% lower validation loss compared to GPT-2 across multiple model depths.

– ShipGuard AI: AI-Based Early Warning System for Shipment Delays

Predicts shipment delay risk up to 62 hours in advance with explainable ML alerts and intervention recommendations.

- * Engineered a full-stack shipment delay early-warning platform using React, FastAPI, XGBoost, and SHAP, achieving 0.913 ROC-AUC and classifying shipments into HIGH / MEDIUM / LOW risk with 0–100% delay probability. Integrated live weather, traffic, port congestion, customs dwell, and news signals, automated Telegram alerts every 5 minutes for shipments crossing 65% risk, and built ranked intervention recommendations with INR cost delta and ROI forecasting to reduce SLA breach risk.
- * GitHub: <https://github.com/amanmurari/ShipGuard-AI/>

– nanoKimi: Lightweight LLM Research Pipeline

Inspired by nanoGPT and Kimi-K2 ideas for efficient training and inference experimentation.

- * Built a compact and reproducible LLM research pipeline integrating Muon optimizer, Mixture of Experts, and Latent Attention.
- * Benchmarked nanoKimi against nanoGPT on training speed, inference throughput, model quality, and parameter efficiency.
- * GitHub: <https://github.com/amanmurari/nanokimi/>

– Multimodal Retrieval-Augmented Generation System

Text and image retrieval system using vector search and multimodal AI workflows.

- * Built a multimodal RAG system for text and image retrieval using FAISS-based similarity search.
- * Designed retrieval pipelines for document search, image search, and AI-powered question answering across multiple data formats.
- * Improved retrieval speed and answer relevance by combining embedding-based search with structured data processing.

– Reinforcement Learning Autonomous Traffic Control System

Reward-based traffic signal control simulation using reinforcement learning.

- * Developed a reinforcement learning-based traffic-control environment where agents learn signal-control strategies through feedback.
- * Designed custom reward functions, traffic scenarios, state-action logic, and policy improvement flow to optimize traffic movement.
- * Applied reinforcement learning concepts such as environment design, reward shaping, policy learning, and feedback-based improvement.

OPEN SOURCE

– Open Source Contribution: nano-GPT-OSS

Efficient Transformer research project focused on reproducible LLM experiments.

- * Core contributor to nano-GPT-OSS, which reached 100+ GitHub stars and 30+ forks.
- * Worked on efficient attention mechanisms, MoE-enhanced model blocks, distributed training, and reproducible evaluation pipelines.
- * Published implementation details, training logs, and evaluation metrics to support open research and community learning.

RESEARCH

– Efficient Transformer Architecture Research

Focused on improving small-scale language model efficiency through attention and MLP architecture changes.

- * Explored hybrid attention designs combining sliding-window attention and sink tokens for efficient context processing.
- * Studied MoE-enhanced MLP layers for improving model capacity without large increases in memory usage.
- * Compared architecture variants against GPT-2 baselines using validation loss, RAM usage, disk footprint, and scaling behavior.

ACHIEVEMENTS & ACTIVITIES

- * Won **1st place** at AI/ML Hackathon Synchronize 4.0, competing against **100+ teams**.
- * Won **1st place** at the **Vizuara AI Research Challenge**, a national-level hackathon with **100+ teams**, earning a two-month paid **AI Research Internship**.
- * Selected as a **finalist** at UIDAI Data Hackathon 2026, a national-level hackathon with **1,000+ teams**.
- * Solved **150+** problems on LeetCode.
- * Core contributor to **nano-GPT-OSS**, an open-source LLM research project with **100+ GitHub stars** and **30+ forks**.
- * Built **Datagentz**, an AI-powered Excel add-in published on Microsoft Marketplace.
- * Worked on efficient Transformer research involving hybrid attention, sink tokens, MoE-enhanced MLPs, and distributed training.
- * Built AI projects across LLMs, RAG, reinforcement learning, multimodal retrieval, and time-series forecasting.

CERTIFICATIONS

- * TensorFlow Bootcamp
- * Data Structures and Algorithms in Java - Apna College
- * Multi-Agent AI Systems Certification

SKILLS

- **Programming Languages:** Python, JavaScript, Rust, PHP, HTML, CSS
- **AI, ML & Data:** Machine Learning, Deep Learning, Reinforcement Learning, NLP, Computer Vision, Generative AI, Pandas, NumPy, Scikit-learn, XGBoost
- **LLMs & GenAI:** Transformers, LLMs, SLMs, RAG, Fine-tuning, RL Fine-tuning, Multi-Agent Systems, Prompt Engineering
- **Model Architectures:** CNN, RNN, LSTM, GAN, Seq2Seq, Attention, MoE, MHA, GQA, MLA
- **Reinforcement Learning:** PPO, DQN, GRPO, reward design, environment design, policy optimization
- **Frameworks & Libraries:** PyTorch, TensorFlow, FastAPI, Django, OpenCV, LangChain, CrewAI, AutoGen
- **Databases & Tools:** PostgreSQL, MySQL, Pinecone, Weaviate, Docker, Git, GitHub, Vector Databases
- **Data Visualization:** Matplotlib, Seaborn, Plotly

LANGUAGES

English, Hindi